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FUJISAKI et al.(10) **Pub. No.: US 2025/0283457 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **FLUID CONTROL DEVICE**(71) Applicant: **Murata Manufacturing Co., Ltd.**,
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Hiroki ACHIWA, Kyoto (JP)(21) Appl. No.: **19/071,037**(22) Filed: **Mar. 5, 2025**(30) **Foreign Application Priority Data**

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ABSTRACT

A fluid control device includes: a pump including a pump housing; and a valve including a valve housing, the pump and the valve being disposed side by side in a first direction. The pump has a first hole formed in the pump housing, a second hole formed in the pump housing, and a vibrator disposed in the pump housing, the vibrator being configured to be able to send gas from the first hole to the second hole. The valve has the second hole formed in a first valve housing member of the valve housing, the first valve housing member being connected to the pump housing, a third hole formed in a second valve housing member of the valve housing, the second valve housing member being disposed to face the first valve housing member, and a discharge passage formed in the second valve housing member of the valve housing.

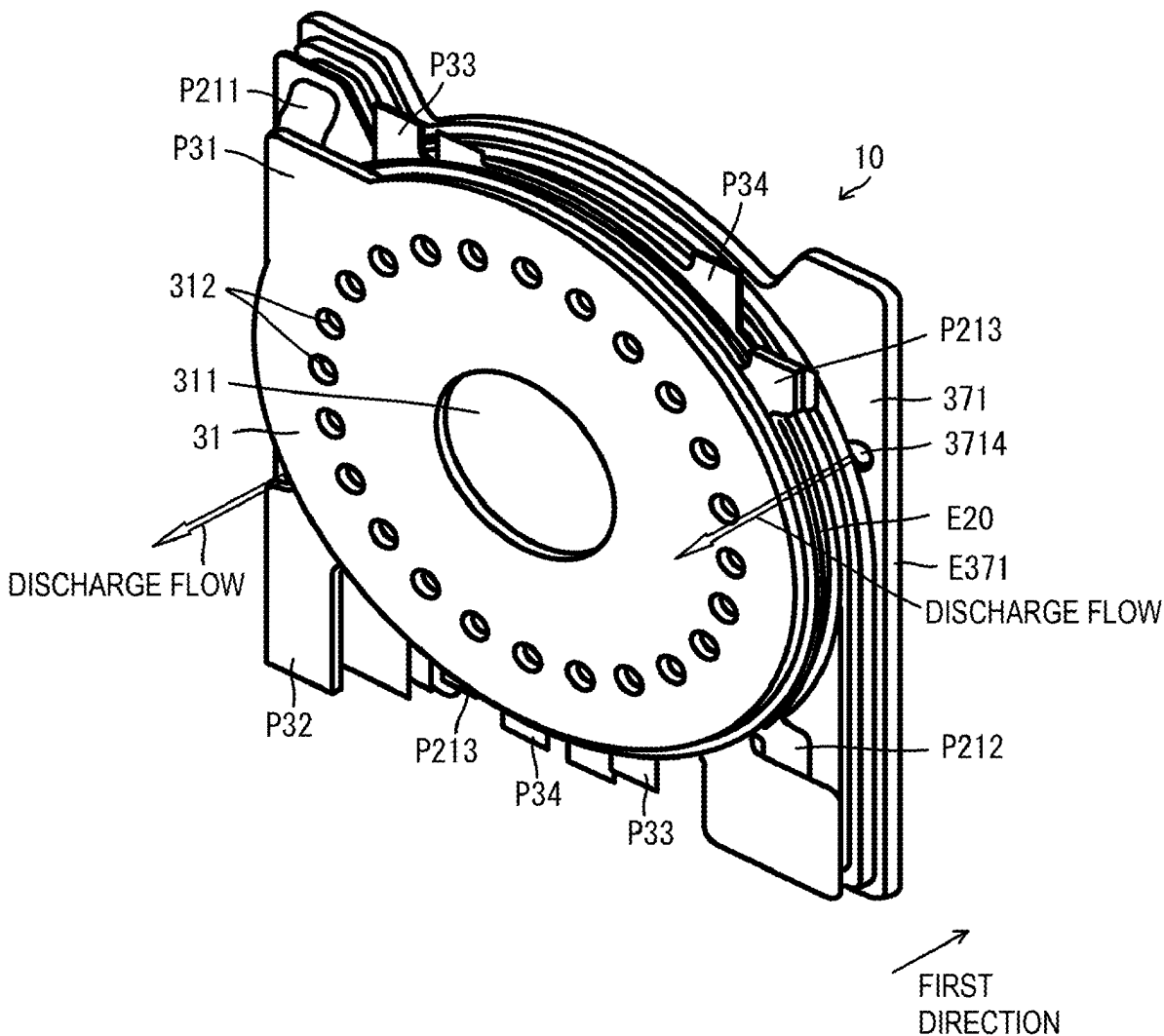


FIG. 1

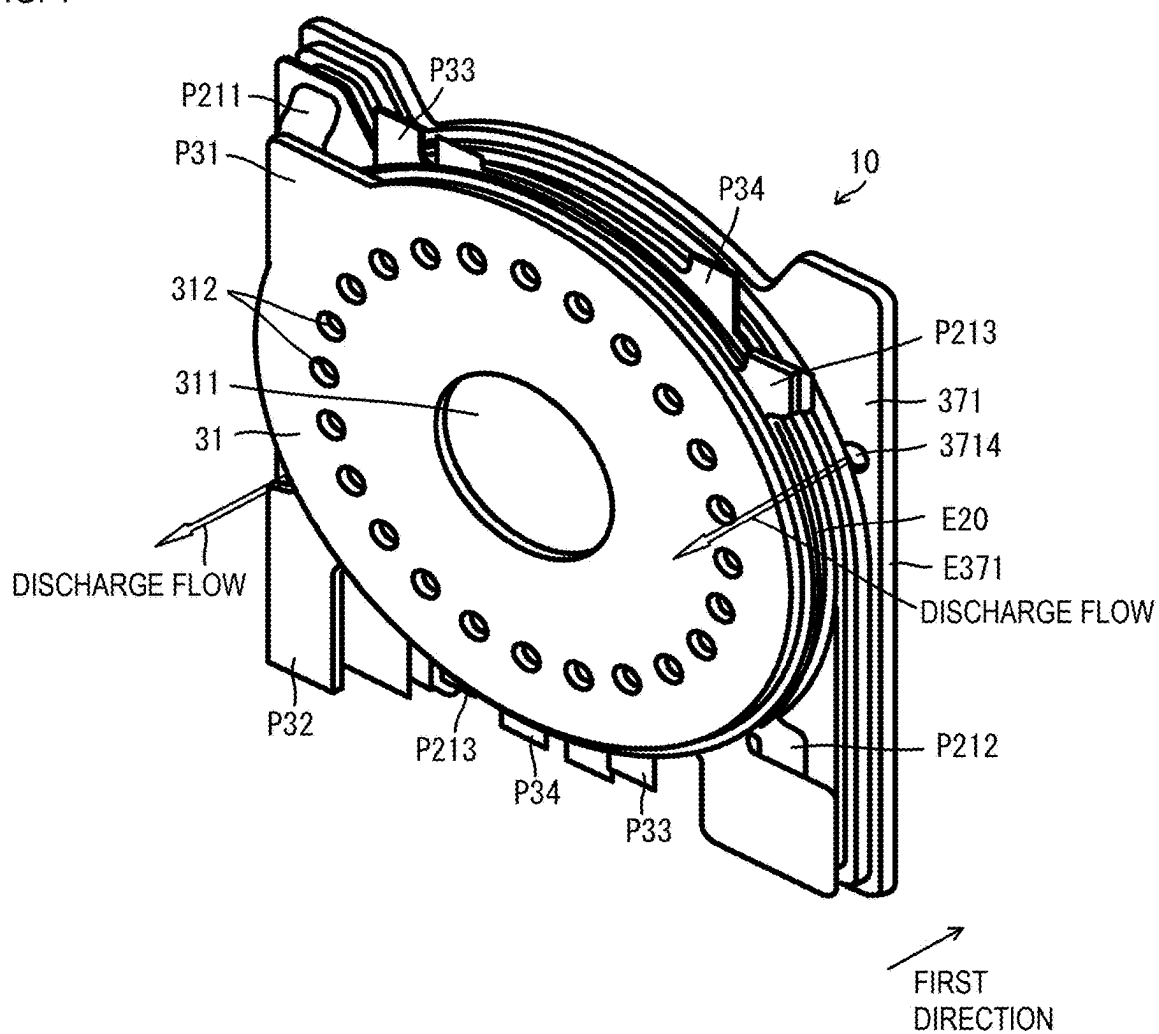


FIG. 2

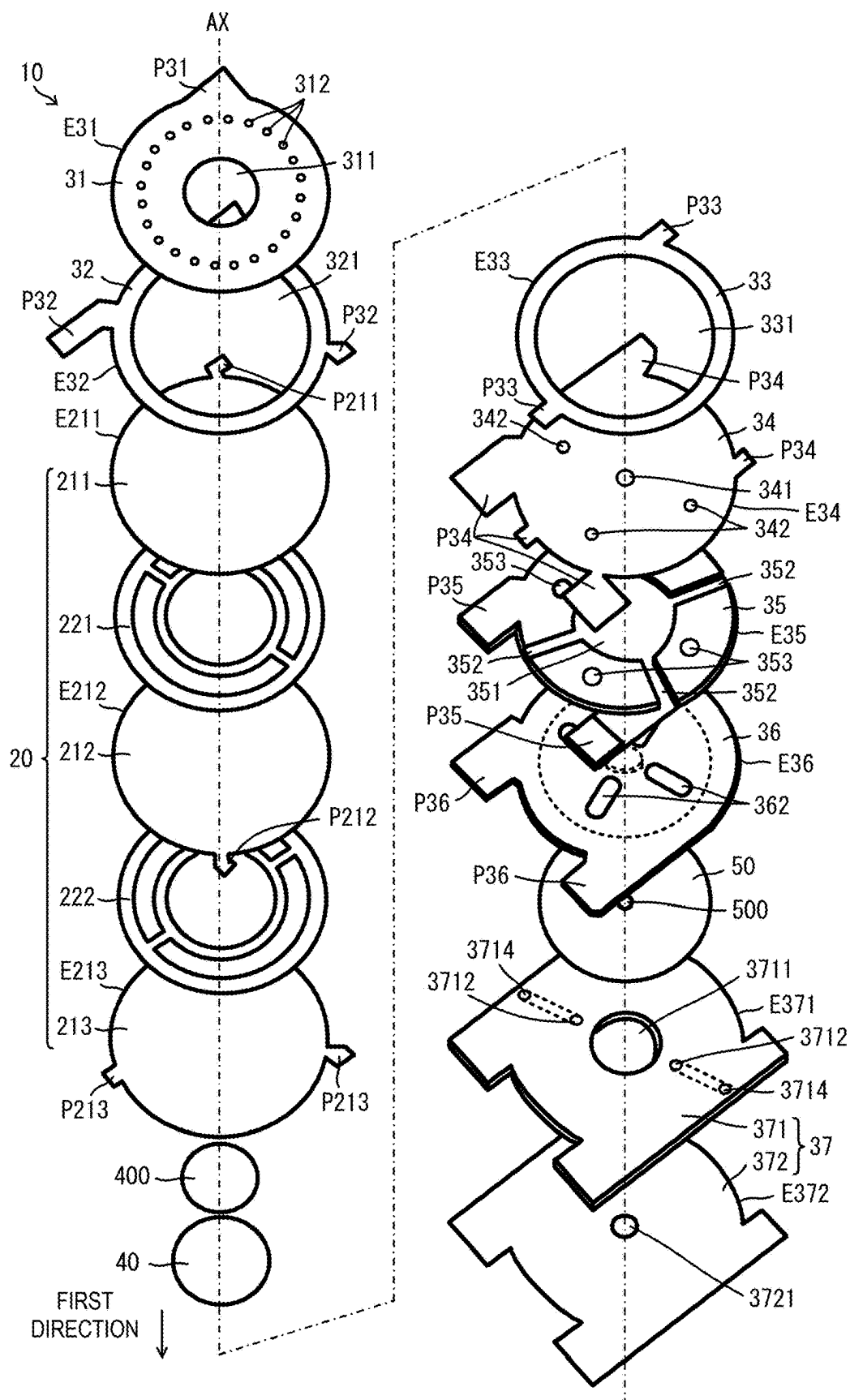


FIG. 3

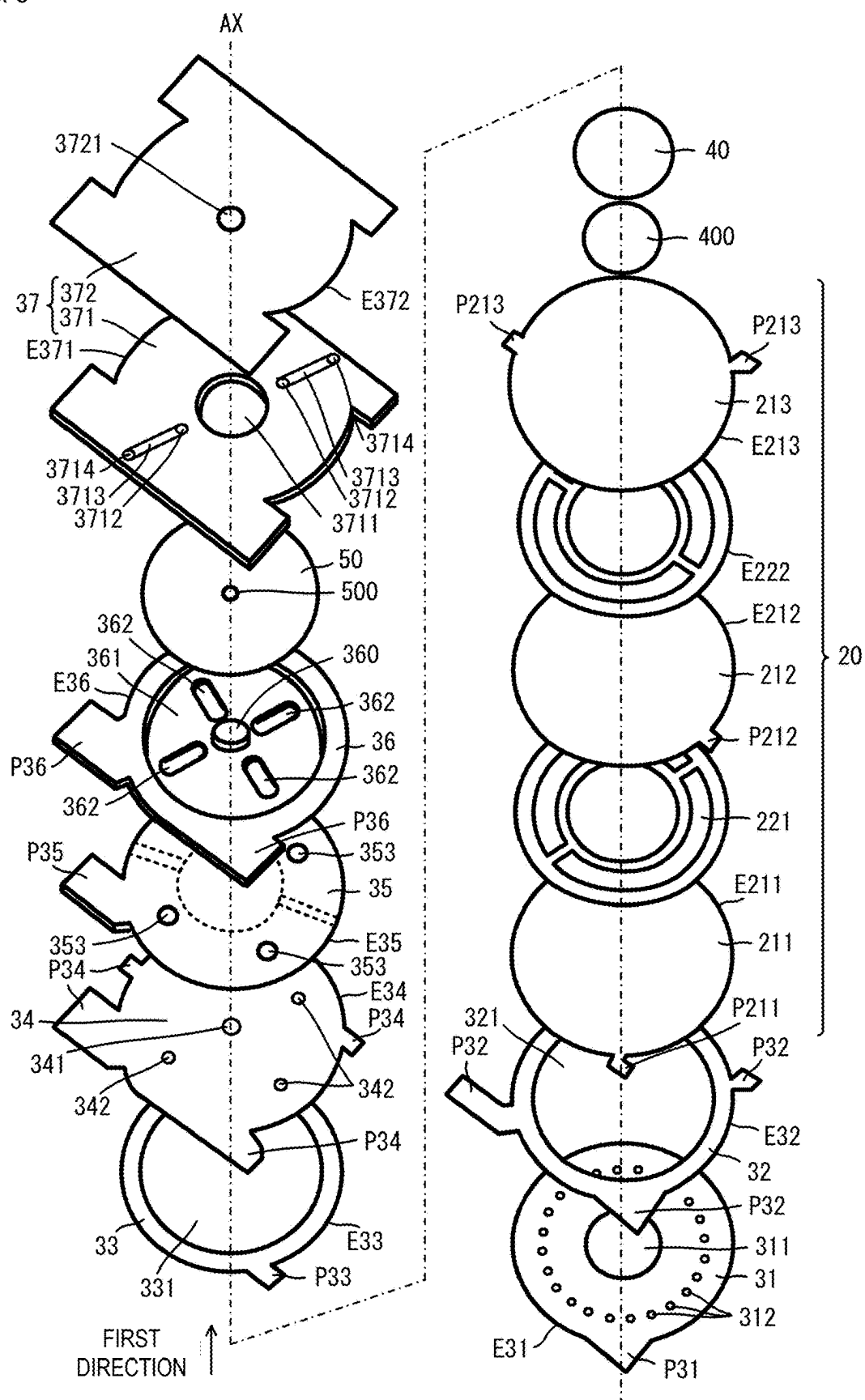


FIG. 4

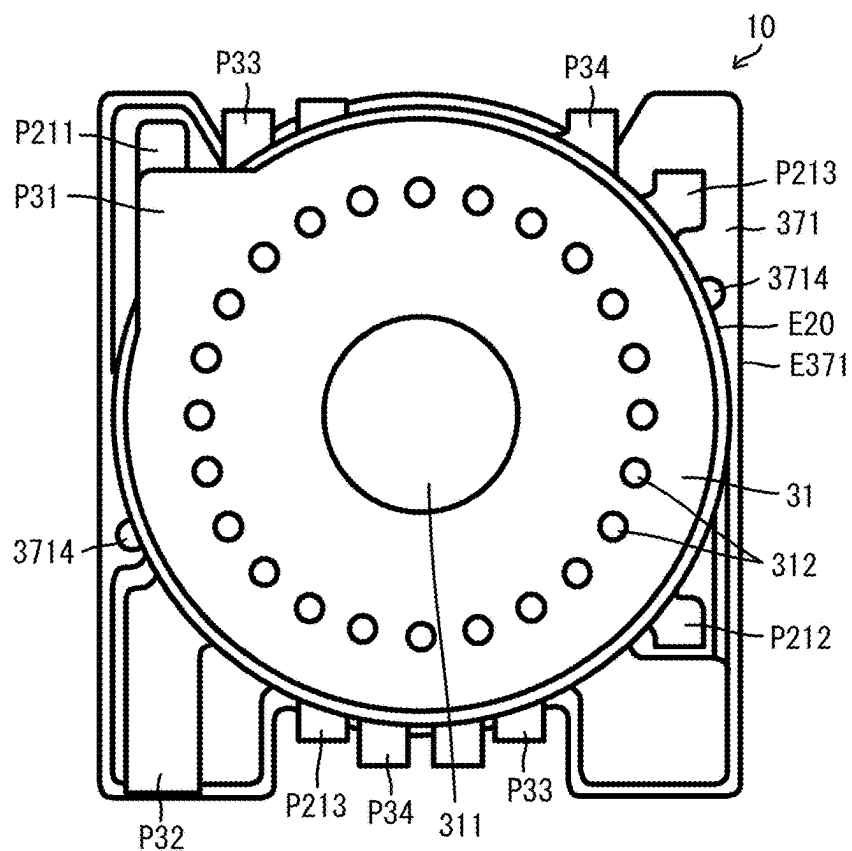
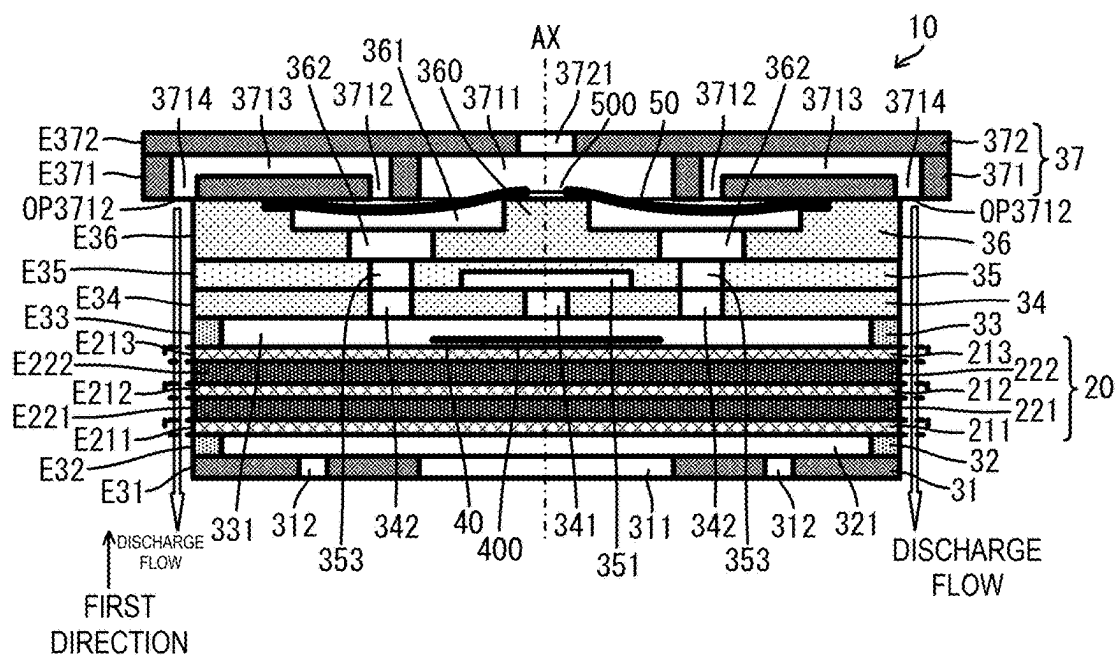


FIG. 5



FLUID CONTROL DEVICE

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims priority from Japanese Patent Application No. 2024-032865 filed on Mar. 5, 2024. The content of this application is incorporated herein by reference in its entirety.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

[0002] The present disclosure relates to a fluid control device configured to convey a fluid such as gas.

2. Description of the Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2016-200067 describes a fluid control device. The fluid control device described in Japanese Unexamined Patent Application Publication No. 2016-200067 includes a pump housing, a rapid discharge valve, and a heat sink. The rapid discharge valve includes an ejection hole and a discharge passage.

[0004] The heat sink is physically connected to the pump housing. Part of the heat sink is disposed at a position facing the discharge passage.

[0005] When driving a pump, the fluid control device supplies air to a cuff communicating with the ejection hole of the rapid discharge valve. When stopping driving the pump, the fluid control device discharges the air collected in the cuff from the discharge passage of the rapid discharge valve.

[0006] The air discharged from the discharge passage comes in contact with the heat sink and cools the heat sink, thus cooling the pump housing.

BRIEF SUMMARY OF THE DISCLOSURE

[0007] However, in such an existing fluid control device described in Japanese Unexamined Patent Application Publication No. 2016-200067, cooling of a pump housing requires a heat sink. Thus, the fluid control device is increased in size. In addition, the pump housing is indirectly cooled. Thus, it is difficult for the fluid control device to achieve high cooling efficiency.

[0008] Accordingly, a possible benefit of the present disclosure is to provide a small fluid control device having high cooling efficiency.

[0009] A fluid control device according to an embodiment of the present disclosure includes: a pump including a pump housing; and a valve including a valve housing, the pump and the valve being disposed side by side in a first direction. The pump has a first hole formed in the pump housing, a second hole formed in the pump housing, and a vibrator disposed in the pump housing, the vibrator being configured to be able to send gas from the first hole to the second hole. The valve has the second hole formed in a first valve housing member of the valve housing, the first valve housing member being connected to the pump housing, a third hole formed in a second valve housing member of the valve housing, the second valve housing member being disposed to face the first valve housing member, and a discharge passage formed in the second valve housing member of the valve housing.

[0010] The second valve housing member of the valve housing includes a projection formed by a part not overlapping an external shape of the pump housing when viewed in the first direction. The projection has at least part of the discharge passage. An outside end surface of the discharge passage opened to an outside faces the pump in the first direction. The vibrator includes a vibrator projection projecting from the pump housing when viewed in the first direction. At least part of the outside end surface of the discharge passage is located outside the pump housing when viewed in the first direction.

[0011] With this configuration, the gas discharged from the discharge passage of the valve flows along an outer side surface of the pump housing. That is, a discharge flow along the outer side surface of the pump housing is generated. Thus, the pump housing is directly cooled by the discharge flow without a heat sink, for example. In addition, the discharged gas flows also in the vicinity of the vibrator projection. Thus, the vibrator projection is directly cooled by the discharge flow, thus cooling the vibrator in the pump housing.

[0012] According to the present disclosure, it is possible to form a small fluid control device having high cooling efficiency.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0013] FIG. 1 is an external perspective view of a fluid control device according to an embodiment of the present disclosure;

[0014] FIG. 2 is an exploded perspective view of the fluid control device according to the embodiment of the present disclosure;

[0015] FIG. 3 is an exploded perspective view of the fluid control device according to the embodiment of the present disclosure;

[0016] FIG. 4 is a plan view of the fluid control device according to the embodiment of the present disclosure when viewed in a first direction; and

[0017] FIG. 5 is a side sectional view illustrating the configuration of the fluid control device according to the embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0018] A fluid control device according to an embodiment of the present disclosure will be described with reference to the drawings. FIG. 1 is an external perspective view of a fluid control device according to an embodiment of the present disclosure. Each of FIGS. 2 and 3 is an exploded perspective view of the fluid control device according to the embodiment of the present disclosure. FIG. 4 is a plan view of the fluid control device according to the embodiment of the present disclosure when viewed in a first direction. FIG. 5 is a side sectional view illustrating the configuration of the fluid control device according to the embodiment of the present disclosure. FIG. 5 is a schematic side sectional view for facilitating understanding of the configuration of the fluid control device.

[0019] A fluid control device 10 includes a vibrator 20, a plurality of housing members 31 to 37, a membrane 40, a membrane 50, and a membrane fixing member 400. The housing members 34 and 35 correspond to housing members

forming a pump housing and correspond to part of a first valve housing member. The housing member 36 corresponds to part of the first valve housing member. The housing member 37 corresponds to a second valve housing member. In the fluid control device 10, an aspect in which housing members common to the pump housing and a valve housing are used is illustrated. However, housing members forming the pump housing and housing members forming the valve housing may be separate from each other.

[0020] The vibrator 20 includes a plurality of flat plates 211, 212, and 213 and a plurality of piezoelectric bodies 221 and 222. The plurality of flat plates 211, 212, and 213 are made of a metal and are configured to be able to deform (vibrate) by being subjected to a stress generated by distortion of the plurality of piezoelectric bodies 221 and 222.

[0021] The flat plate 211 has an outer peripheral end E211 having a substantially circular shape in plan view. The flat plate 211 includes a projection P211 projecting outward from the outer peripheral end E211. For example, the projection P211 has a substantially rectangular shape in plan view.

[0022] The flat plate 212 has an outer peripheral end E212 having a substantially circular shape in plan view. The flat plate 212 includes a projection P212 projecting outward from the outer peripheral end E212. For example, the projection P212 has a substantially rectangular shape in plan view.

[0023] The flat plate 213 has an outer peripheral end E213 having a substantially circular shape in plan view. The flat plate 213 includes projections P213 projecting outward from the outer peripheral end E213. For example, the projections P213 have substantially rectangular shapes in plan view.

[0024] The outer peripheral end E211 of the flat plate 211, the outer peripheral end E212 of the flat plate 212, and the outer peripheral end E213 of the flat plate 213 have substantially the same shape. The projection P211, the projection P212, and the projections P213 each correspond to a vibrator projection of the present disclosure.

[0025] The plurality of piezoelectric bodies 221 and 222 have a circular shape in plan view.

[0026] In the vibrator 20, the flat plate 211, the piezoelectric body 221, the flat plate 212, the piezoelectric body 222, and the flat plate 213 are laminated in this order in the first direction. The flat plate 211, the piezoelectric body 221, the flat plate 212, the piezoelectric body 222, and the flat plate 213 are laminated such that the respective centers of the flat plate 211, the piezoelectric body 221, the flat plate 212, the piezoelectric body 222, and the flat plate 213 substantially coincide with each other when viewed in the first direction. Thus, the vibrator 20 forms a bimorph actuator.

[0027] In this case, the outer peripheral ends E211, E212, and E213 of the plurality of flat plates 211, 212, and 213 substantially coincide with each other when viewed in the first direction. Then, the position where the outer peripheral ends E211, E212, and E213 overlap each other corresponds to an outer peripheral end E20 of the vibrator 20.

[0028] The housing member 31 is a flat plate having an outer peripheral end E31 having a substantially circular shape in plan view. The outer peripheral end E31 and the outer peripheral end E20 of the vibrator 20 have substantially the same shape. The housing member 31 includes a projection P31 projecting outward from the outer peripheral end E31.

[0029] The housing member 31 has a through hole 311 and a plurality of through holes 312. The through hole 311 and the plurality of through holes 312 have circular shapes when viewed from the side where a surface where each through hole is open is located. The through hole 311 is formed at a position including the center of the housing member 31 in plan view. The center of the through hole 311 substantially coincides with the center of a circle formed by the outer peripheral end E31. The plurality of through holes 312 are formed between the through hole 311 and the outer peripheral end E31 at intervals of a substantially equal angle in a circumferential direction surrounding the through hole 311.

[0030] The housing member 32 is a circular ring-shaped body having an outer peripheral end E32 having a substantially circular shape in plan view. The outer peripheral end E32 and the outer peripheral end E31 have substantially the same shape. The housing member 32 includes projections P32 projecting outward from the outer peripheral end E32.

[0031] The housing member 32 is a ring-shaped body having a through hole 321. The diameter of the through hole 321 is larger than the diameter of a circle formed by the plurality of through holes 312 of the housing member 31.

[0032] The housing member 33 is a circular ring-shaped body having an outer peripheral end E33 having a substantially circular shape in plan view. The outer peripheral end E33 and the outer peripheral ends E31 and E32 have substantially the same shape. The housing member 33 includes projections P33 projecting outward from the outer peripheral end E33.

[0033] The housing member 33 has a through hole 331 as a space inside the ring-shaped body. The diameter of the through hole 331 is substantially equal to the diameter of the through hole 321 of the housing member 32.

[0034] The housing member 34 is a flat plate having an outer peripheral end E34 having a substantially circular shape in plan view. The outer peripheral end E34 and the outer peripheral end E33 have substantially the same shape. The housing member 34 includes projections P34 projecting outward from the outer peripheral end E34.

[0035] The housing member 34 has a through hole 341 and a plurality of through holes 342. The through hole 341 and the plurality of through holes 342 have circular shapes when viewed from the side where a surface where each through hole is open is located. The through hole 341 is formed at a position including the center of the housing member 34 in plan view. The center of the through hole 341 substantially coincides with the center of a circle formed by the outer peripheral end E34. The plurality of through holes 342 are circumferentially formed between the through hole 341 and the outer peripheral end E34.

[0036] The housing member 35 is a flat plate having an outer peripheral end E35 having a substantially circular shape in plan view. The outer peripheral end E35 and the outer peripheral end E34 have substantially the same shape. The housing member 35 includes projections P35 projecting outward from the outer peripheral end E35.

[0037] The housing member 35 has a recess 351, a plurality of grooves 352, and a plurality of through holes 353. The recess 351 has a shape recessed from one main surface of the housing member 35. The recess 351 has a circular shape when viewed from the side where a surface where the recess 351 is open is located. The recess 351 is formed at a position including the center of the housing member 35 in

plan view. The center of the recess 351 substantially coincides with the center of a circle formed by the outer peripheral end E35.

[0038] As with the recess 351, the plurality of grooves 352 each have a shape recessed from the one main surface of the housing member 35. The plurality of grooves 352 each have a belt shape. The plurality of grooves 352 are formed radially from the recess 351 to the outer peripheral end E35. The plurality of grooves 352 are formed at intervals of a predetermined angle in a circumferential direction surrounding the recess 351. The plurality of grooves 352 reach the outer peripheral end E35 and are open to the outside from the outer peripheral end E35.

[0039] The plurality of through holes 353 have a circular shape when viewed from the side where a surface where each through hole is open is located. The plurality of through holes 353 are circumferentially formed between the recess 351 and the outer peripheral end E35. The plurality of through holes 353 are formed between the plurality of grooves 352 in the circumferential direction surrounding the recess 351.

[0040] The housing member 36 is a flat plate having an outer peripheral end E36 having a substantially circular shape in plan view. The outer peripheral end E36 and the outer peripheral end E35 have substantially the same shape. The housing member 36 includes projections P36 projecting outward from the outer peripheral end E36.

[0041] The housing member 36 includes a recess 361, a plurality of through holes 362, and a seat 360. The recess 361 has a shape recessed from one main surface of the housing member 36. The recess 361 has a circular shape when viewed from the side where a surface where the recess 361 is open is located. The recess 361 is formed at a position including the center of the housing member 36 in plan view. The center of the recess 361 substantially coincides with the center of a circle formed by the outer peripheral end E36.

[0042] The plurality of through holes 362 each have an elliptical shape when viewed from the side where a surface where each through hole is open is located. The plurality of through holes 362 are circumferentially formed around the center of the housing member 36. The plurality of through holes 362 are each formed such that the long-side direction of the elliptical shape is parallel to a direction extending radially from the center of the housing member 36. The plurality of through holes 362 overlap the recess 361 in plan view and communicate with a recess internal space formed by the recess 361 and the membrane 50.

[0043] The seat 360 has a cylindrical shape. The seat 360 has a shape projecting from a bottom surface of the recess 361. The seat 360 is formed at a position including the center of the housing member 36 in plan view.

[0044] The housing member 37 includes a housing member 371 and a housing member 372.

[0045] The housing member 371 is a flat plate having an outer peripheral end E371 having a substantially circular shape in plan view. The shape of the outer peripheral end E371 is larger than the shape of each of the outer peripheral ends E31 to E36. The housing member 371 includes projections projecting outward from the outer peripheral end E371.

[0046] The housing member 371 has a through hole 3711, through holes 3712, grooves 3713, and through holes 3714.

[0047] The through hole 3711 has a circular shape when viewed from the side where a surface where the through hole

is open is located. The through hole 3711 is formed at a position including the center of the housing member 371 in plan view. The center of the through hole 3711 substantially coincides with the center of a circle formed by the outer peripheral end E371.

[0048] The through holes 3712 and 3714 have a circular shape when viewed from the side where the surface where each through hole is open is located. The diameter of the through holes 3712 and 3714 is smaller than the diameter of the through hole 3711. The through holes 3712 and 3714 are formed between the through hole 3711 and the outer peripheral end E371. The through holes 3712 and 3714 are arranged in directions extending radially from the center of the housing member 37. The through holes 3714 are formed at positions closer to the outer peripheral end E371 than the through holes 3712.

[0049] The grooves 3713 each have a shape recessed from one main surface of the housing member 371. The groove 3713 is formed between the through hole 3712 and the through hole 3714 in plan view. The groove 3713 communicates with the through hole 3712 and the through hole 3714.

[0050] Two pairs of the through holes 3712, the grooves 3713, and the through holes 3714 are formed. The pairs are disposed at positions located point-symmetrically relative to the center of the housing member 37.

[0051] The housing member 372 is a flat plate having an outer peripheral end E372 having a substantially circular shape in plan view. The outer peripheral end E372 and the outer peripheral end E371 have the same shape. The housing member 372 includes projections projecting outward from the outer peripheral end E372.

[0052] The housing member 372 has a through hole 3721. The through hole 3721 has a circular shape when viewed from the side where a surface where the through hole is open is located. The through hole 3721 is formed at a position including the center of the housing member 372 in plan view. The center of the through hole 3721 substantially coincides with the center of a circle formed by the outer peripheral end E372. The diameter of the through hole 3721 is smaller than the diameter of the through hole 3711 of the housing member 371.

[0053] The housing member 372 is in contact with a surface of the housing member 371 where the grooves 3713 are open. Thus, the housing member 371 and the housing member 372 have discharge passages formed by the through holes 3712, the grooves 3713 whose openings are closed by the housing member 371, and the through holes 3714. The housing member 371 and the housing member 372 may be formed separately from each other or integrally with each other.

[0054] The membrane 40 is made of a flexible material. The membrane 40 has a circular shape.

[0055] The membrane 50 is made of a flexible material. The membrane 50 has a circular shape. The membrane 50 has a through hole 500. The through hole 500 has a circular shape when viewed from the side where a surface where the through hole is open is located. The diameter of the through hole 500 is smaller than the diameter of the seat 360 of the housing member 36.

[0056] The housing member 31 and the housing member 32 are disposed to be laminated on one side of the vibrator 20 in the first direction. The housing members 33 to 37 are

disposed to be laminated on the other side of the vibrator 20 in the first direction. More specifically, this configuration is as follows.

[0057] The housing member 32 is disposed on the side of the vibrator 20 closer to the flat plate 211 and is connected to the flat plate 211. In this case, the outer peripheral end E32 of the housing member 32 coincides with the outer peripheral end E20 of the vibrator 20 when viewed in the first direction.

[0058] The housing member 31 is disposed on the opposite side of the housing member 32 from the vibrator 20 and is connected to the housing member 32. In this case, the outer peripheral end E31 of the housing member 31 coincides with the outer peripheral end E32 of the housing member 32 when viewed in the first direction.

[0059] The housing member 33 is disposed on the side of the vibrator 20 closer to the flat plate 213 and is connected to the flat plate 213. In this case, the outer peripheral end E33 of the housing member 33 coincides with the outer peripheral end E20 of the vibrator 20 when viewed in the first direction.

[0060] The housing member 34 is disposed on the opposite side of the housing member 33 from the vibrator 20 and is connected to the housing member 33. In this case, the outer peripheral end E34 of the housing member 34 coincides with the outer peripheral end E33 of the housing member 33 when viewed in the first direction.

[0061] The through hole 341 and the plurality of through holes 342 of the housing member 34 communicate with the through hole 331 of the housing member 33.

[0062] The housing member 35 is disposed on the opposite side of the housing member 34 from the housing member 33 and is connected to the housing member 34. In this case, the housing member 35 is connected to the housing member 34 such that the side of the housing member 35 where the recess 351 and the plurality of grooves 352 are open is closer to the housing member 34. In addition, the outer peripheral end E35 of the housing member 35 coincides with the outer peripheral end E34 of the housing member 34 when viewed in the first direction.

[0063] The recess 351 and the plurality of grooves 352 of the housing member 35 communicate with the through hole 341 of the housing member 34. The plurality of through holes 353 of the housing member 35 communicate with the plurality of respective through holes 342 of the housing member 34.

[0064] The housing member 36 is disposed on the opposite side of the housing member 35 from the housing member 34 and is connected to the housing member 35. In this case, the housing member 36 is connected to the housing member 35 such that the opposite side of the housing member 36 from the side where the recess 361 is open is closer to the housing member 35. In addition, the outer peripheral end E36 of the housing member 36 coincides with the outer peripheral end E35 of the housing member 35 when viewed in the first direction.

[0065] The plurality of through holes 362 of the housing member 36 communicate with the plurality of respective through holes 353 of the housing member 35.

[0066] The housing member 371 is disposed on the opposite side of the housing member 36 from the housing member 35 and is connected to the housing member 36. In this case, the outer peripheral end E371 of the housing member 371 forming the housing member 37 and the outer

peripheral end E372 of the housing member 372 forming the housing member 37 coincide with the outer peripheral end E36 of the housing member 36 when viewed in the first direction.

[0067] The through hole 3711 and the through holes 3712 of the housing member 37 communicate with the recess 361 of the housing member 36. The through holes 3714 of the housing member 37 are located outside the outer peripheral end E36 of the housing member 36. In the housing member 37, the through hole 3711 and the through hole 3721 communicate with each other.

[0068] The membrane 40 is disposed in a space surrounded by the vibrator 20, the housing member 33, and the housing member 34. The membrane 40 is fixed with the membrane fixing member 400 to cover the central position of the flat plate 213. A central portion of the membrane 40 is fixed, and a peripheral portion of the membrane 40 is movably fixed.

[0069] The membrane 50 is disposed in a space (the recess 361) surrounded by the housing member 36 and the housing member 37. The membrane 50 is fixed by holding an outer peripheral portion of the membrane 50 between the housing member 36 and the housing member 37. A central portion of the membrane 50 is movably fixed. The through hole 500 of the membrane 50 overlaps the seat 360 when viewed in the first direction.

[0070] With this configuration, when a driving voltage is applied to the vibrator 20, the vibrator 20 generates bending vibrations. The application of a driving voltage to the vibrator 20 is performed by using the flat plates 211, 212, and 213, for example. In this case, it is possible to use each projection.

[0071] Bending vibrations of the vibrator 20 vary the volume of the space (pump space) surrounded by the vibrator 20, the housing member 33, and the housing member 34. Thus, the gas outside the fluid control device 10 is suctioned into the space (pump space) surrounded by the vibrator 20, the housing member 33, and the housing member 34 from the grooves 352 through the recess 351 and the through hole 341.

[0072] The gas suctioned into the space surrounded by the vibrator 20, the housing member 33, and the housing member 34 flows into the space (valve space) surrounded by the housing member 36 and the housing member 37 through the through holes 342, the through holes 353, and the through holes 362.

[0073] In this manner, the fluid control device 10 includes the pump housing formed by the plurality of housing members 31 to 36, and the vibrator 20 and thus functions as a pump. Then, the grooves 352 each correspond to a suction hole (first hole) of the fluid control device 10, and the through holes 342, the through holes 353, and the through holes 362 each correspond to a communication hole (second hole) of the fluid control device 10.

[0074] In this case, the membrane 40 is deformed by a gas flow. Thus, the membrane 40 does not close a passage when gas is suctioned into the pump space from the grooves 352 and closes the passage when gas flows into the pump space from the through holes 342. Accordingly, the fluid control device 10 can reduce the backflow of gas.

[0075] The gas that has flowed into the space (valve space) surrounded by the housing member 36 and the housing member 37 pushes up the membrane 50, and the membrane 50 is then separated from a seat 360. Thus, the gas flows into

the through hole 3711 and the through hole 3721 of the housing member 37 through the through hole 500 of the membrane 50 and is ejected to the outside of the fluid control device 10 from the through hole 3721. That is, the through hole 3721 corresponds to an ejection hole (third hole) of the fluid control device 10.

[0076] In this case, the membrane 50 closes the through holes 3712. Thus, the fluid control device 10 can inhibit the leakage of gas through the through holes 3712, the grooves 3713, and the through holes 3714.

[0077] When the application of a driving voltage to the vibrator 20 is stopped, bending vibrations of the vibrator 20 are stopped. In this case, for example, when a cuff (not illustrated) is set to the through hole 3721, gas flows backward from the cuff through the through hole 3721. The backflow of the gas pushes down the membrane 50, and the membrane 50 comes in contact with the seat 360. Thus, the through hole 500 of the membrane 50 is closed by the seat 360. Accordingly, the fluid control device 10 can inhibit the back-flown gas from flowing backward toward the pump.

[0078] Then, the movement of the membrane 50 toward the seat 360 opens the through holes 3712. Thus, the back-flown gas is discharged through the through holes 3712, the grooves 3713, and the through holes 3714. That is, the through holes 3712, the grooves 3713, and the through holes 3714 correspond to the discharge passages of the fluid control device 10.

[0079] In this manner, the fluid control device 10 includes the valve housing formed by the plurality of housing members 36 and 37, and the membrane 50 and thus functions as a valve.

[0080] In this configuration, the shape of an outer peripheral end E37 of the housing member 37 is larger than the shape of each of the outer peripheral ends E31 to E36 of the other housing members 31 to 36 forming the fluid control device 10. Thus, the through holes 3714 forming the discharge passages are located outside an outer peripheral end of the pump housing.

[0081] In addition, the through holes 3714 are open toward the housing members 31 to 36. That is, the end surfaces (outside end surfaces) OP3712 of the discharge passages exposed to the outside are exposed to face the pump housing.

[0082] Thus, the gas discharged through each discharge passage (discharge flow) flows toward the pump housing (see discharge flows represented by thick arrows in FIGS. 1 and 5). The gas discharged from each discharge passage flows along an outer side surface (outer surface formed by the outer peripheral ends E31 to E36 of the plurality of housing members 31 to 36) of the pump housing. That is, a discharge flow along the outer side surface of the pump housing is generated. Thus, the pump housing is directly cooled by the discharge flow without a heat sink, for example. Accordingly, the fluid control device 10 can achieve high cooling efficiency with a small size.

[0083] In addition, when viewed in the first direction, the plurality of projections P211, P212, and P213 of the vibrator 20 do not overlap the end surfaces OP3712 of the discharge passages exposed to the outside. Thus, part of the discharged gas is not blocked by the pump housing and the plurality of projections P211, P212, and P213 of the vibrator 20, and the gas flow is not obstructed. In addition, it is possible to inhibit the discharge flow from being disturbed by the plurality of

projections P211, P212, and P213. Accordingly, the fluid control device 10 can achieve high cooling efficiency.

[0084] When viewed in the first direction, the projections of the plurality of housing members 31 to 36 also do not overlap the end surfaces OP3712 of the discharge passages exposed to the outside. Thus, it is possible to inhibit the discharge flow from being blocked and disturbed by the projections of the plurality of housing members 31 to 36. Accordingly, the fluid control device 10 can achieve high cooling efficiency.

[0085] In addition, when viewed in the first direction, the end surfaces OP3712 of the discharge passages exposed to the outside may be as close to the outer peripheral end of the pump housing (outer peripheral ends E31 to E36 of the plurality of housing members 31 to 36) as possible. Furthermore, when viewed in the first direction, the end surfaces OP3712 of the discharge passages exposed to the outside may have parts overlapping the outer peripheral end of the pump housing. Thus, the effect of cooling by the discharged gas is further improved.

[0086] In addition, the number of the discharge passages may be two or more. This increases the number of parts of the pump housing to be cooled, thus improving the cooling effect. For example, in the present embodiment, two discharge passages are used to further improve the cooling effect than that in the case of having one discharge passage. Furthermore, three or more discharge passages can improve the cooling effect.

[0087] In addition, the plurality of discharge passages may be disposed at positions located point-symmetrically relative to the center of the valve housing when viewed in the first direction. This reduces variations in parts to be cooled in a circumferential direction, thus improving the cooling effect.

[0088] In addition, in the fluid control device 10, the plurality of flat plates 211, 212, and 213 respectively include the projections P211, P212, and P213. The projections P211, P212, and P213 project outward from the pump housing. Accordingly, the discharged gas flows also in the vicinity of each of the projections P211, P212, and P213. As a result, the projections P211, P212, and P213 are directly cooled by the discharge flows, thus cooling the vibrator 20 in the pump housing.

[0089] In this case, the flat plate 211 including the projection P211, the flat plate 212 including the projection P212, and the flat plate 213 including the projections P213 are made of a metal and thus have a high thermal conductivity. Accordingly, the fluid control device 10 can achieve a higher cooling effect.

[0090] <1> A fluid control device comprising: a pump including a pump housing; and a valve including a valve housing, the pump and the valve being disposed side by side in a first direction, wherein the pump has a first hole formed in the pump housing, a second hole formed in the pump housing, and a vibrator disposed in the pump housing, the vibrator being configured to be able to send gas from the first hole to the second hole, the valve has the second hole formed in a first valve housing member of the valve housing, the first valve housing member being connected to the pump housing, a third hole formed in a second valve housing member of the valve housing, the second valve housing member being disposed to face the first valve housing member, and a discharge passage formed in the second valve housing member of the valve housing, the second valve housing member includes a projection formed by a part not overlap-

ping an external shape of the pump housing when viewed in the first direction, the projection has at least part of the discharge passage, an outside end surface of the discharge passage opened to an outside faces the pump in the first direction, the vibrator includes a vibrator projection projecting from the pump housing when viewed in the first direction, and at least part of the outside end surface of the discharge passage is located outside the pump housing when viewed in the first direction.

[0091] <2> The fluid control device according to <1>, wherein the outside end surface of the discharge passage does not overlap the vibrator projection when viewed in the first direction.

[0092] <3> The fluid control device according to <1> or <2>, further comprising a plurality of the discharge passages, wherein the discharge passages are disposed at positions located point-symmetrically relative to a center of the valve housing when viewed in the first direction.

[0093] <4> The fluid control device according to any one of <1> to <3>, wherein the outside end surface of the discharge passage has a part overlapping the pump housing when viewed in the first direction.

[0094] <5> The fluid control device according to any one of <1> to <4>, wherein the vibrator has a multilayer structure including a piezoelectric body and a vibration body made of a metal, and the vibration body is held by the pump housing and includes the vibrator projection.

What is claimed is:

1. A fluid control device comprising:
 - a pump including a pump housing; and
 - a valve including a valve housing, the pump and the valve being disposed side by side in a first direction, wherein the pump has
 - a first hole provided in the pump housing,
 - a second hole provided in the pump housing, and
 - a vibrator disposed in the pump housing, the vibrator being configured to send gas from the first hole to the second hole,
- the valve has
 - the second hole provided in a first valve housing member of the valve housing, the first valve housing member being connected to the pump housing,

- a third hole provided in a second valve housing member of the valve housing, the second valve housing member being disposed to face the first valve housing member, and

- a discharge passage provided in the second valve housing member of the valve housing,

the second valve housing member includes a projection provided by a part not overlapping an external shape of the pump housing when viewed in the first direction, the projection has at least part of the discharge passage, an outside end surface of the discharge passage opened to an outside faces the pump in the first direction, the vibrator includes a vibrator projection projecting from the pump housing when viewed in the first direction, and

at least part of the outside end surface of the discharge passage is located outside the pump housing when viewed in the first direction.

2. The fluid control device according to claim 1, wherein the outside end surface of the discharge passage does not overlap the vibrator projection when viewed in the first direction.

3. The fluid control device according to claim 1, wherein the discharge passage includes a plurality of discharge passages, and

wherein the plurality of discharge passages is disposed at positions located point-symmetrically relative to a center of the valve housing when viewed in the first direction.

4. The fluid control device according to claim 1, wherein the outside end surface of the discharge passage has a part overlapping the pump housing when viewed in the first direction.

5. The fluid control device according to claim 1, wherein the vibrator has a multilayer structure including a piezoelectric body, and a vibration body comprising a metal, and

the vibration body is held by the pump housing and includes the vibrator projection.

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